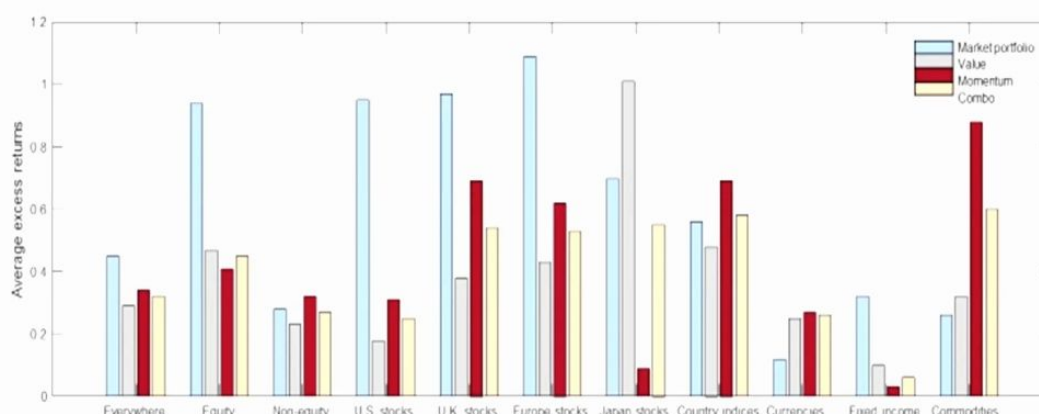


A Global Macroeconomic Risk Model for Value, Momentum, and Other Asset Classes

Overview of Talk

1. Facts: value and momentum across markets and across asset classes
2. What we do and why it is interesting
3. Main results
4. Robustness checks

Empirical Facts: performance of value and momentum strategies across markets and across asset classes (Asness, Moskowitz, and Pedersen, 2013)



What we do

We propose a version of Ross's (1976) Arbitrage Pricing Theory based on a global representation of the Chen, Roll, and Ross's (1986) macroeconomic risk factors:

$$r_{i,t} = \alpha_i + \beta_{i,MP}MP_t + \beta_{i,UI}UI_t + \beta_{i,DEI}DEI_t + \beta_{i,UTS}UTS_t + \beta_{i,URP}UPR_t + \varepsilon_{i,t}$$

- $r_{i,t}$ - return on asset i (or a long-short value or momentum return premium, or a combination of a value and momentum return premia)
- MP_t - industrial production growth
- UI_t - unexpected inflation
- DEI_t - change in expected inflation
- UTS_t - terms spread
- UPR_t - default spread

What We Do

This leads to the following no-arbitrage condition:

$$E(r_{i,t} - r_{f,t}) = \beta_{i,MP}E(r_{MP}) + \beta_{i,UI}E(r_{UI}) + \beta_{i,DEI}E(r_{DEI}) + \beta_{i,UTS}E(r_{UTS}) + \beta_{i,URP}E(r_{URP})$$

where $E(r_{MP})$, $E(r_{UI})$, $E(r_{DEI})$, $E(r_{UTS})$, and $E(r_{URP})$ are the expected returns on the mimicking portfolios for MP , UI , DEI , UTS , and URP , respectively.

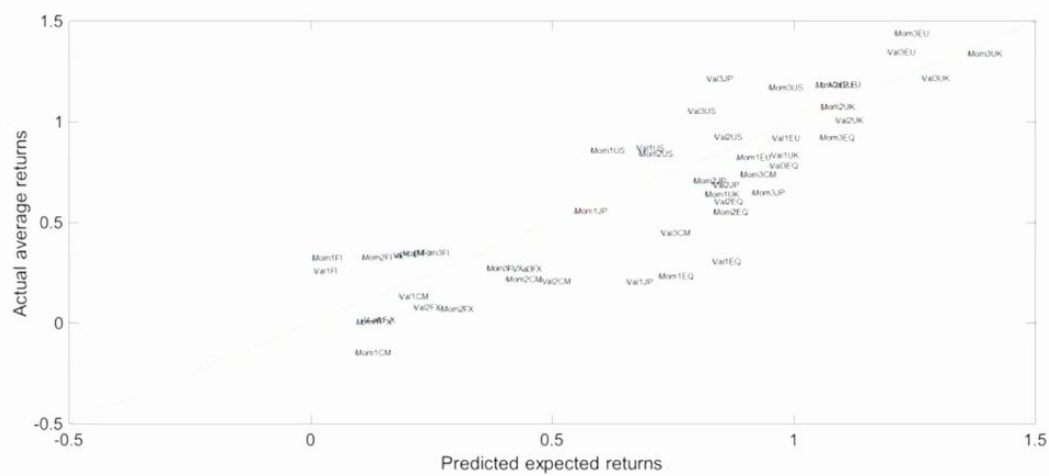
Why It Is Interesting

- Common factor structure across markets and across asset classes
- Economic explanation - global macroeconomic risk - global business cycle
- Differing loadings with respect to the global macroeconomic factors - explain the negative correlation of value and momentum premia
- Asset pricing integration across asset classes and across markets

Tests of Global Integration Across Markets and Across Asset Classes

Model	Global Tangency			Global CRR			Local CRR		
	a	GRS	p(GRS)	a	GRS	p(GRS)	a	GRS	p(GRS)
U.S. stocks	0.20	0.59	0.74	0.20	1.15	0.33	0.19	1.10	0.36
U.K. stocks	0.06	0.32	0.93	0.08	0.40	0.88	0.07	0.45	0.84
Europe stocks	0.14	1.09	0.37	0.12	1.28	0.26	0.13	1.33	0.24
Japan stocks	0.23	2.60	0.02	0.23	3.13	0.01	0.22	2.97	0.01
Country indices	0.31	2.13	0.05	0.31	3.17	0.00			
Currencies	0.11	0.86	0.53	0.13	0.89	0.51			
Fixed income	0.21	2.08	0.05	0.20	2.71	0.01			
Commodities	0.19	0.42	0.86	0.20	0.37	0.90			

Main Results - realized returns vs. expected returns



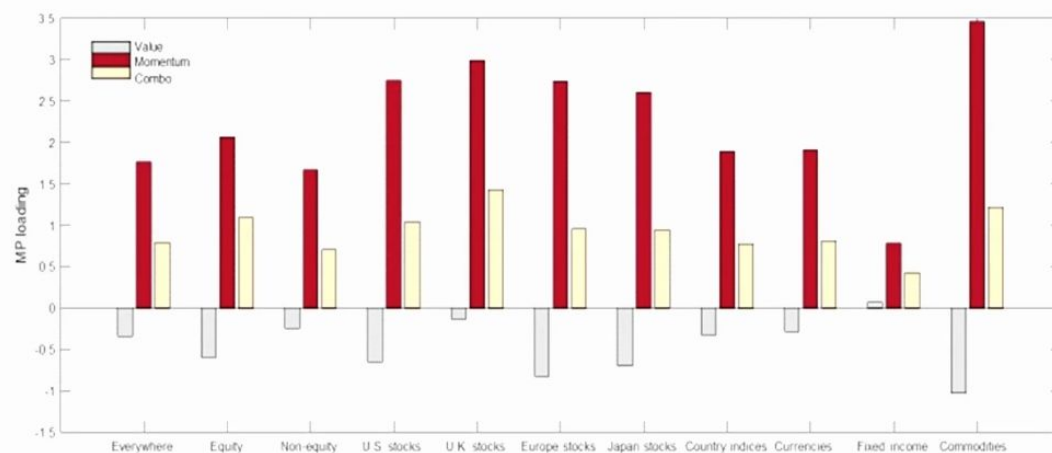
Main Results - Summary statistics of model performance

Model	GRS	p(GRS)	HJ	Diff HJ	HJ^2	$A a_i $	$A a_i / A \tilde{r}_i $	$A \hat{a}_i^2 / A \tilde{r}_i^2$	$A \hat{a}_i^2(a_i) / A \hat{a}_i^2$	AR^2
Global CAPM	3.99	0.000	0.816	0.1980	0.2453	0.57	0.33	0.50	0.39	
AMP	3.99	0.000	0.750	0.0946	0.1848	0.43	0.21	0.85	0.43	
Global CRR	2.82	0.000	0.684		0.1824	0.43	0.18	1.06	0.44	

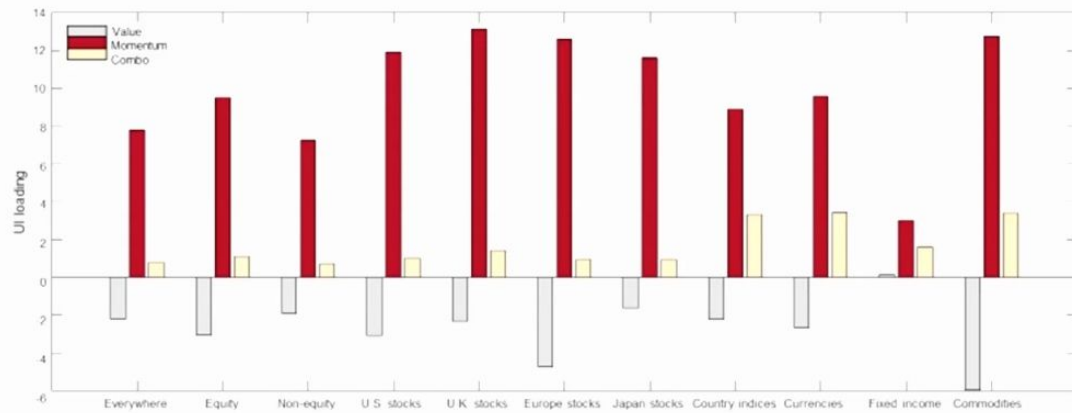
Factor Regressions - Barillas and Shanken (2017)

Factors	MSCI global alphas	Global value and momentum alphas	Global CRR alphas
MSCI global			-0.00 (-0.03)
Global value			0.01 (1.16)
Global momentum			0.01 (0.99)
Global value and momentum			0.01 p(GRS)=0.97 GRS=0.001
MP	-0.06 (-0.62)	0.18 (1.76)	
UI	-0.06 (-1.56)	0.02 (0.52)	
DEI	0.21 (3.33)	-0.13 (-1.77)	
UTS	1.37 (5.09)	-0.09 (-0.40)	
UPR	0.51 (1.18)	0.57 (1.35)	
Global CRR	0.44 p(GRS)=0.00 GRS= 10.81	0.20 p(GRS)=0.00 GRS=1.64	

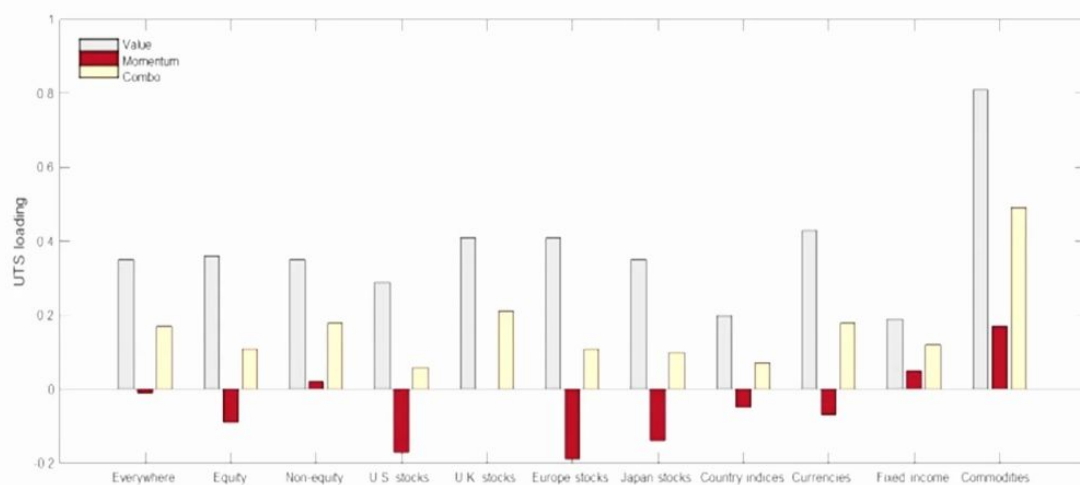
Differing Factor Loadings – value and momentum long-short premia and the combination of the two – Industrial production growth (MP)



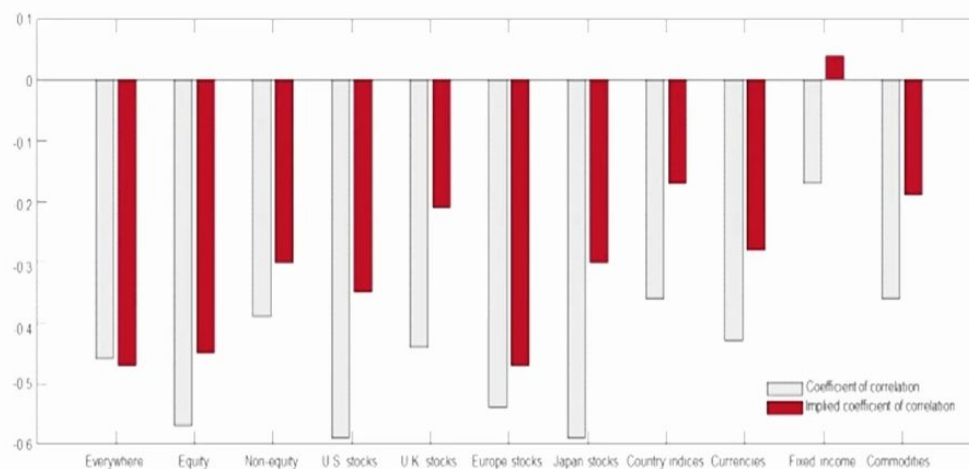
Differing Factor Loadings – value and momentum long-short premia and the combination of the two – Unexpected inflation (UI)



Differing Factor Loadings – value and momentum long-short premia and the combination of the two – Term spread (UTS)



Actual Correlation vs. Implied Correlation of Value and Momentum Premia



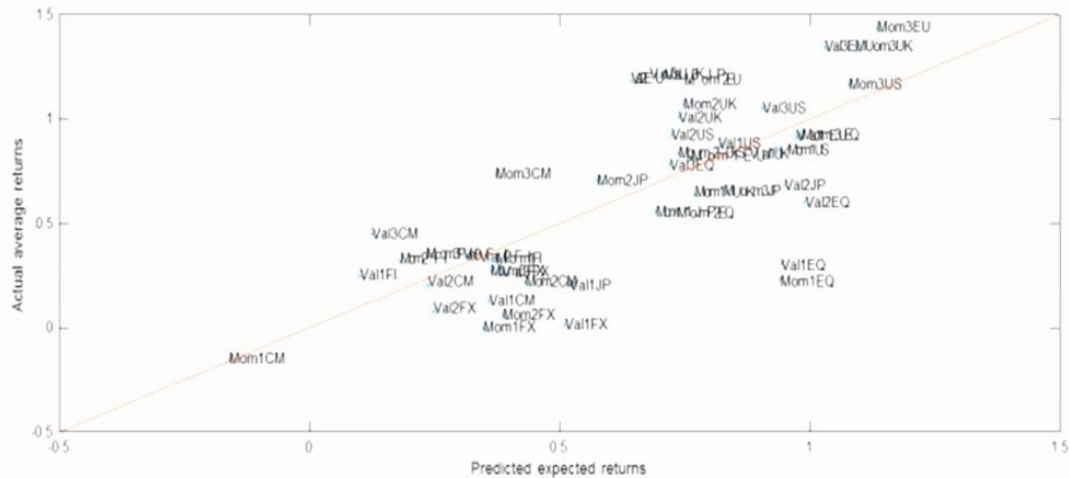
Robustness Checks - Good States and Bad States

Risk premium estimates from two-stage Fama and MacBeth (1973) cross-sectional regressions – 48 value and momentum portfolios

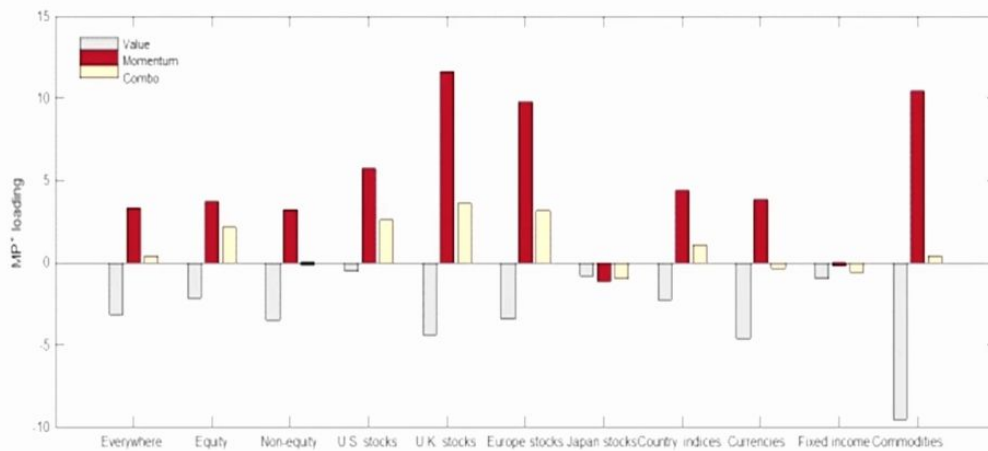
	$\hat{\gamma}_0$	$\hat{\gamma}_{MP+}$	$\hat{\gamma}_{UI+}$	$\hat{\gamma}_{DEI+}$	$\hat{\gamma}_{UTS+}$	$\hat{\gamma}_{UPR+}$	$\hat{\gamma}_{MP-}$	$\hat{\gamma}_{UI-}$	$\hat{\gamma}_{DEI-}$	$\hat{\gamma}_{UTS-}$	$\hat{\gamma}_{UPR-}$
$\hat{\gamma}$	0.21	-0.003	0.005	-0.001	-0.011	-0.10	0.03	0.001	-0.03	0.20	-0.05
	(2.87)	(-0.20)	(2.19)	(-1.16)	(-0.10)	(-3.34)	(1.44)	(0.32)	(-1.1)	(2.37)	(-0.47)
R^2	0.94										
RMSE	0.28										

Robustness Checks - Good States and Bad States

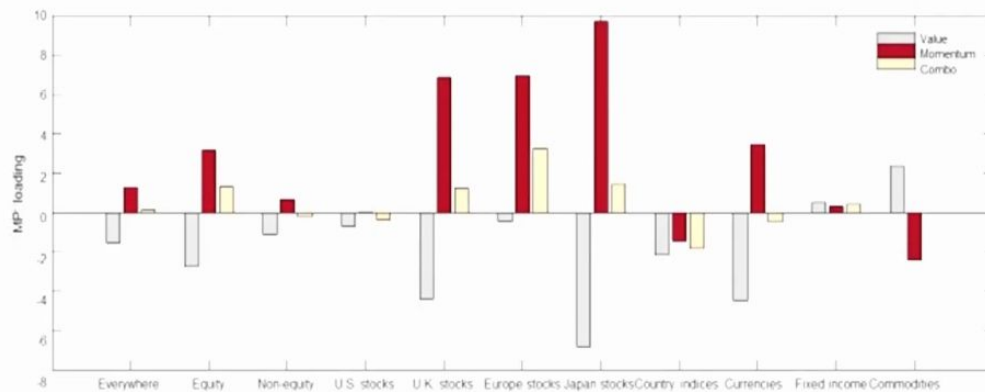
Pricing errors – 48 value and momentum portfolios



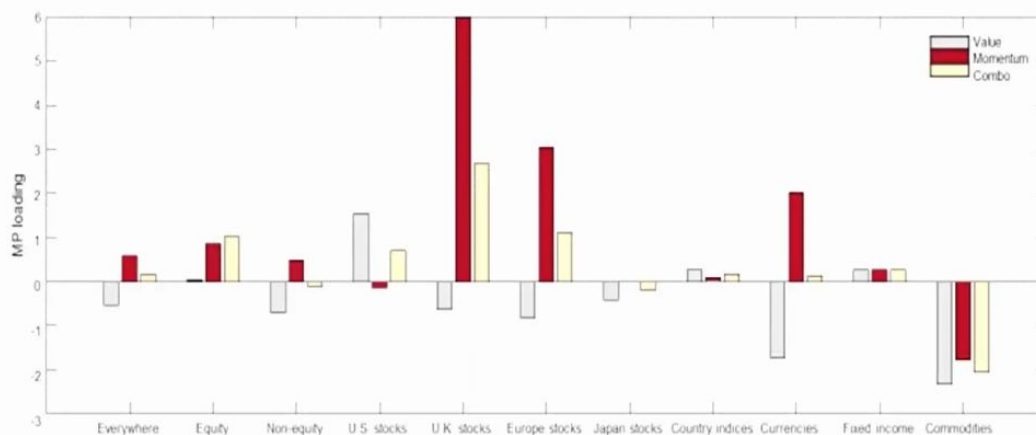
Differing Factor Loadings – value and momentum long-short premia and the combination of the two – Industrial production growth (MP) – Good states



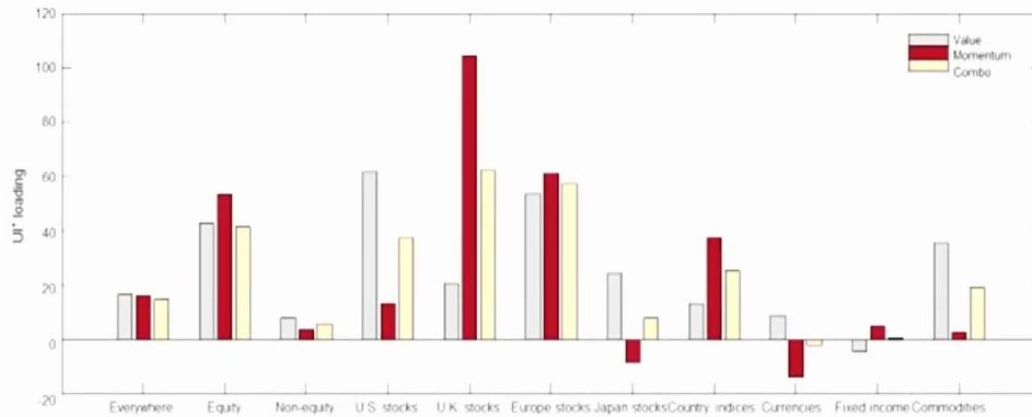
Differing Factor Loadings – value and momentum long-short premia and the combination of the two – Industrial production growth (MP) – Bad states



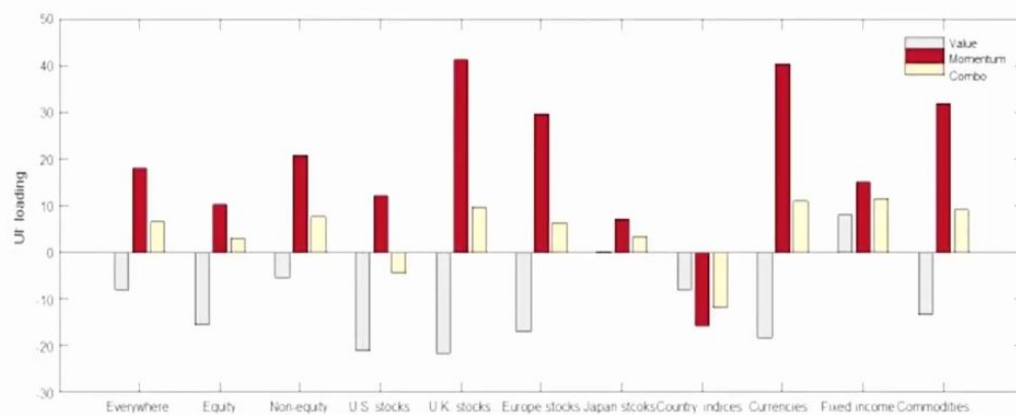
Differing Factor Loadings – value and momentum long-short premia and the combination of the two – Industrial production growth (MP) – Full sample



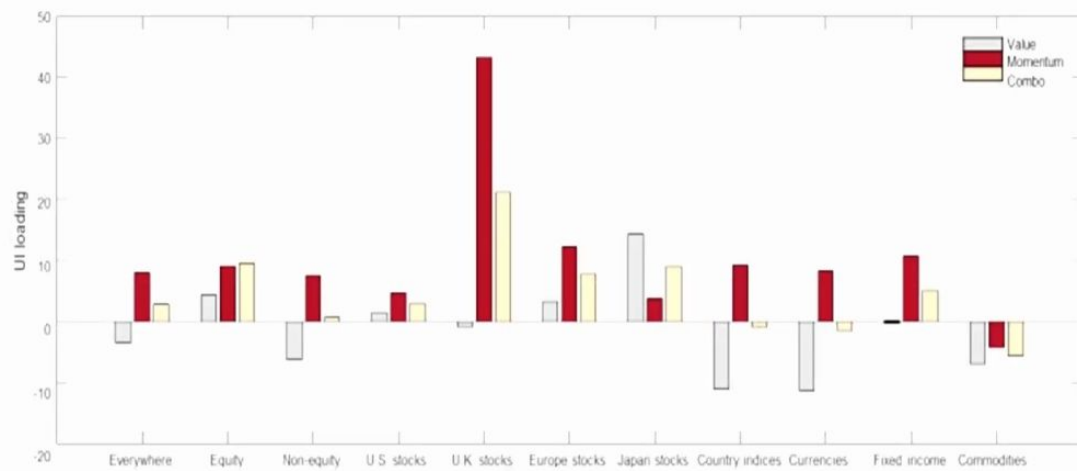
Differing Factor Loadings – value and momentum long-short premia and the combination of the two – Unexpected inflation (UI) – Good states



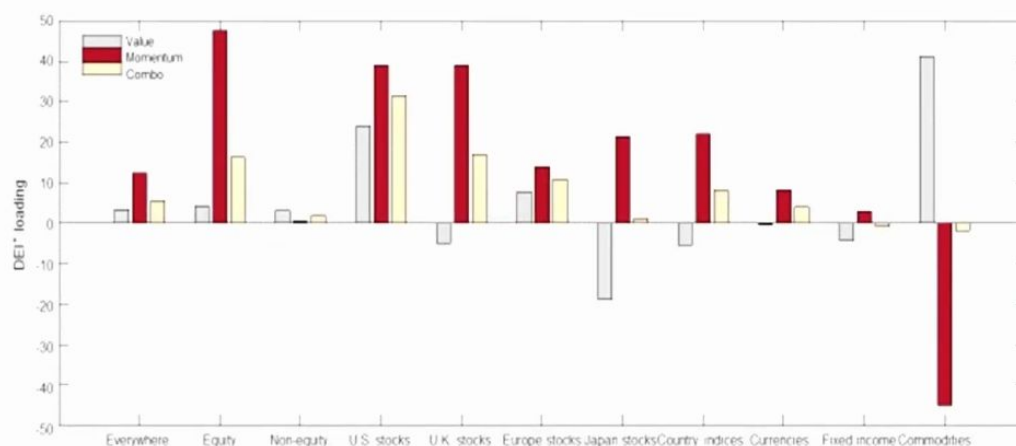
Differing Factor Loadings – value and momentum long-short premia and the combination of the two – Unexpected inflation (UI) – Bad states



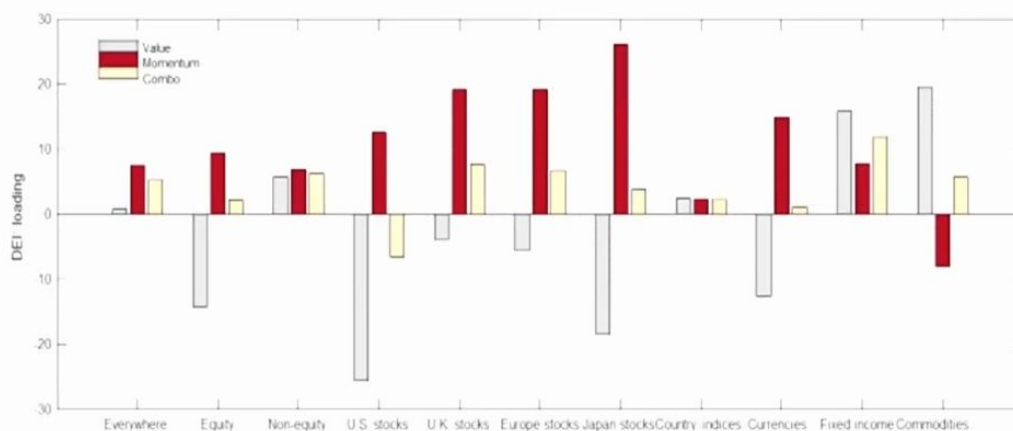
Differing Factor Loadings – value and momentum long-short premia and the combination of the two – Unexpected inflation (UI) – Full sample



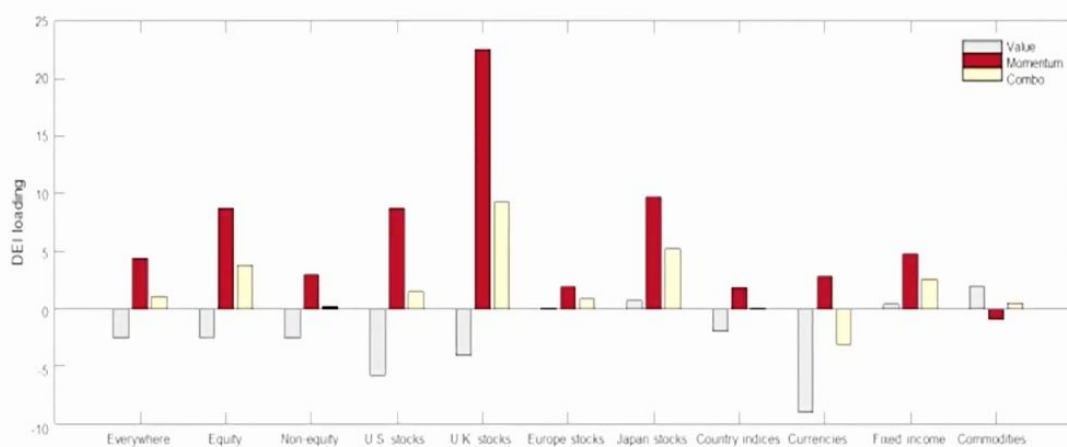
Differing Factor Loadings – value and momentum long-short premia and the combination of the two – Change in expected inflation (DEI) – Good states



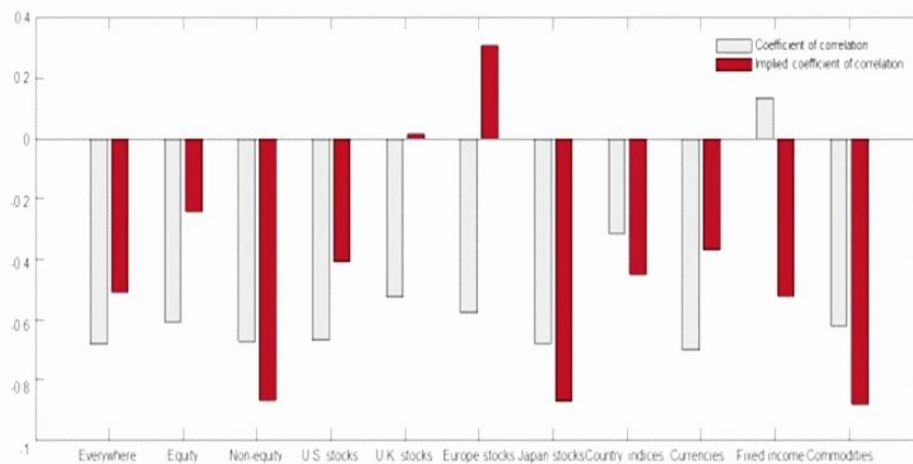
Differing Factor Loadings – value and momentum long-short premia and the combination of the two – Change in expected inflation (DEI) – Bad states



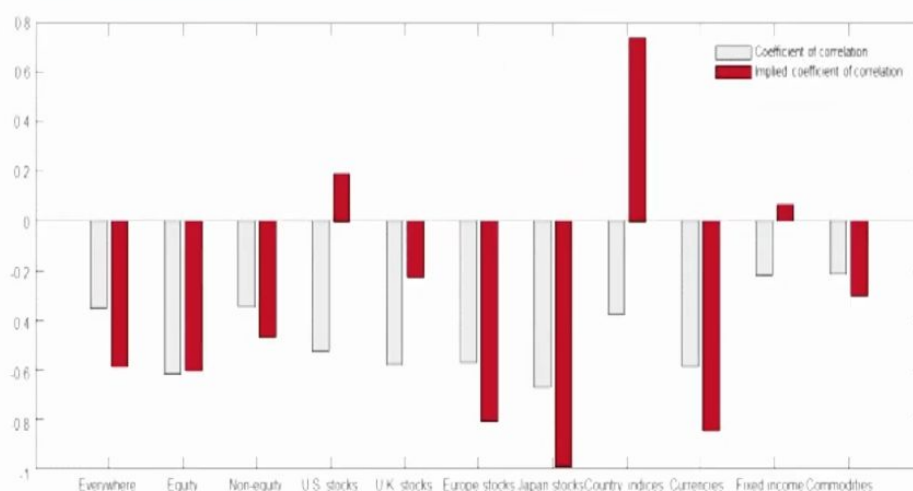
Differing Factor Loadings – value and momentum long-short premia and the combination of the two – Change in expected inflation (DEI) – Full sample



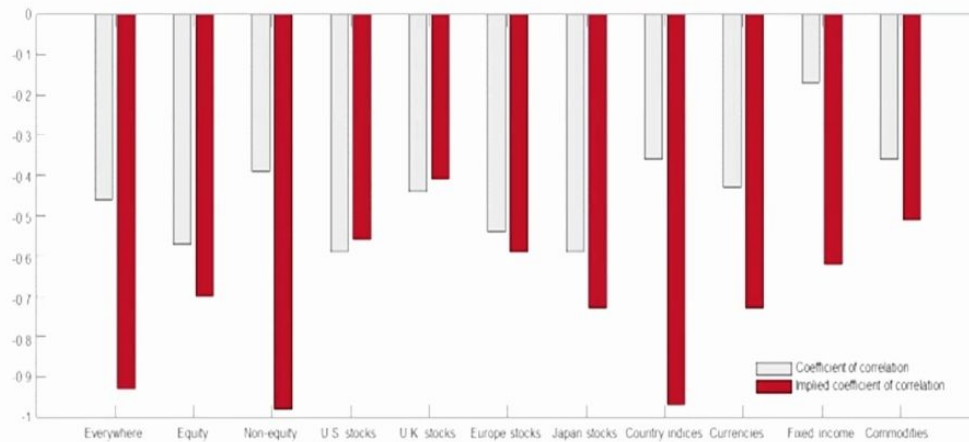
Actual Correlation vs. Implied Correlation of Value and Momentum Premia – Good states



Actual Correlation vs. Implied Correlation of Value and Momentum Premia – Bad states

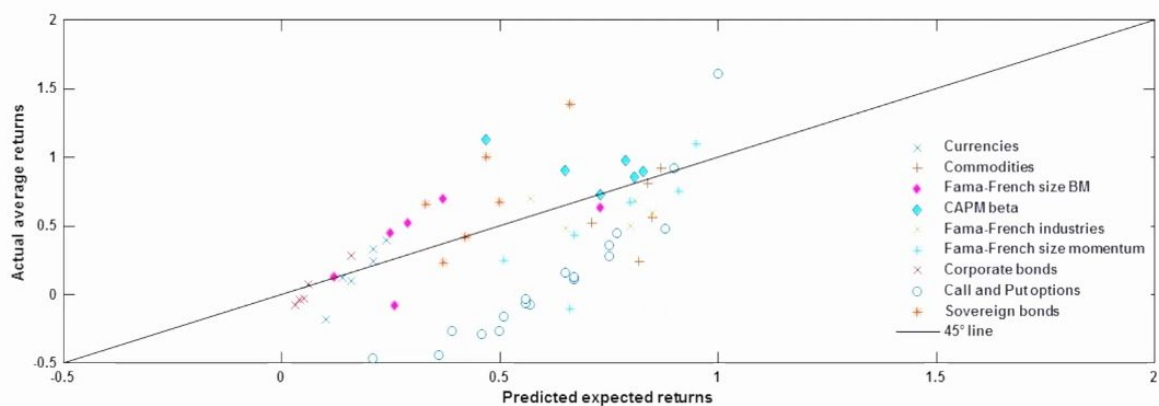


Actual Correlation vs. Implied Correlation of Value and Momentum Premia – Full sample



Robustness Checks – Additional test assets

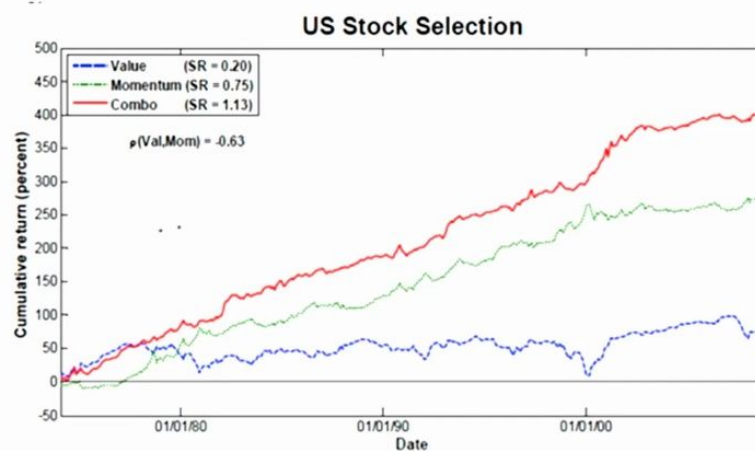
- Describe the average returns on a different set of assets (Lettau, Maggiori, and Weber, 2014)



Holy Grail of Empirical Asset Pricing

- Explain profitability of Value and Momentum strategies around the world, across asset classes
- Relate expected returns to sources of fundamental macroeconomic risk

Combining value and momentum

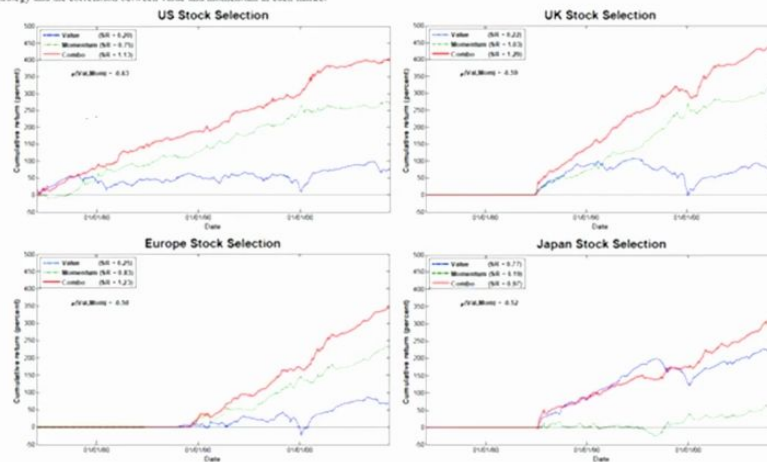


Asness, Moskowitz, and Pedersen (2013)

Value and momentum in global equities

Figure 1: Performance of value and momentum strategies

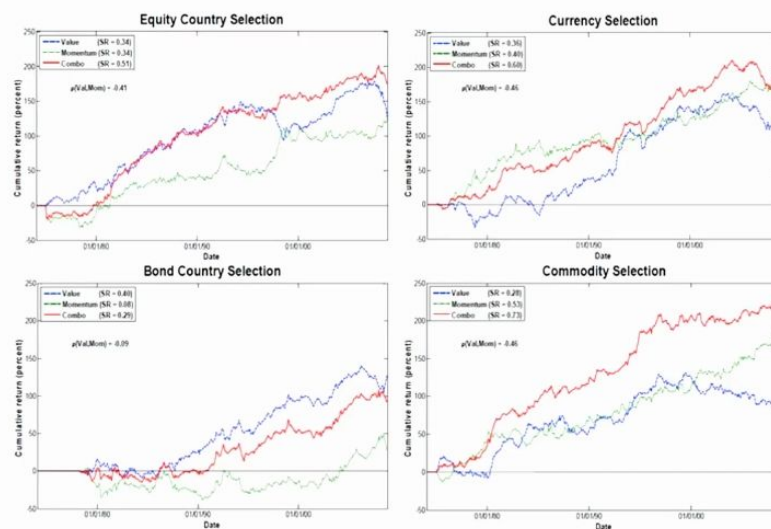
Plotted are the cumulative returns to value, momentum, and a 50/50 combination of value and momentum strategies among individual stocks in four markets: U.S., U.K., Japan, and Continental Europe, in four different asset classes: Country equity index futures, country bonds, currencies, and commodities, and for the equal-weighted combination of all stock selection strategies, all non-stock selection strategies, and an equal-weighted combination of both. Also reported on each figure are the annualized Sharpe ratios of each strategy and the correlation between value and momentum in each market.



Asness, Moskowitz, and Pedersen (2013)

4

Value and momentum “everywhere”



Asness, Moskowitz, and Pedersen (2013)

5

This Paper

- Consider value and momentum across asset classes, globally
- Relate average returns to macro factors of Chen, Roll, and Ross (1986)
- Use factor-mimicking portfolios composed using 6 “global” portfolios formed on value and momentum across assets

Steve Ross and APT: Origins of Factor Investing

- Systematic sources of risk cannot be diversified away
- Hence investors need to be compensated for exposures to them
- Covariances with common factors should explain risk premia

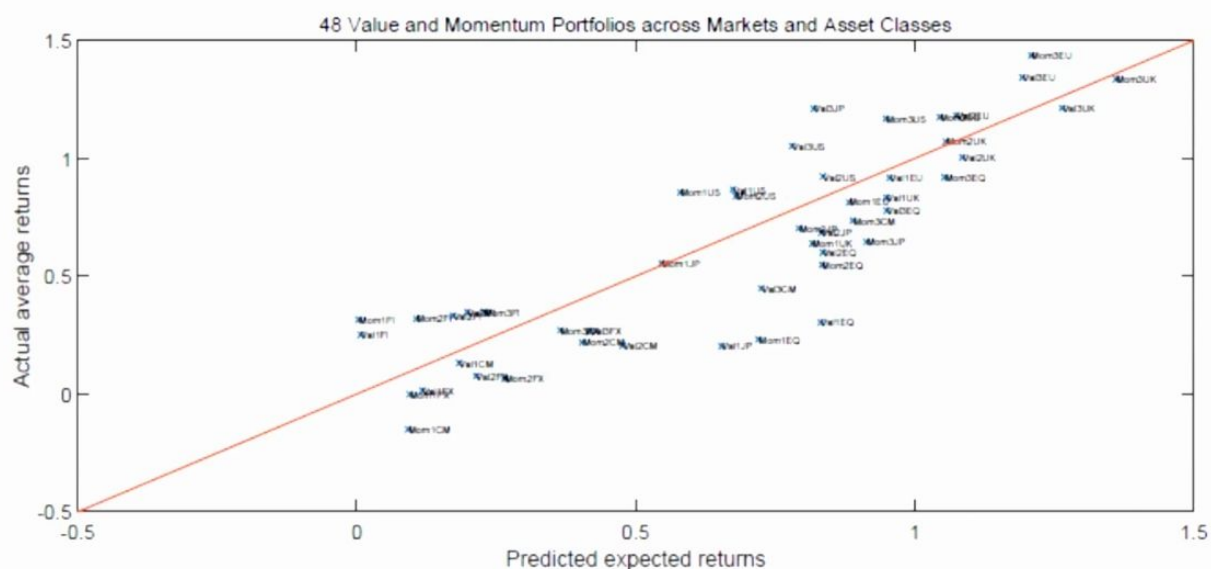
Common Factors Within Asset Classes

- **Bonds**
Litterman and Scheinkman (1991)
- **Stocks**
Fama and French (1992), Carhart (1997)
- **FX**
Lustig, Roussanov, and Verdelhan (2011)
- **Commodities**
Szymanowska, De Roon, Nijman, and Goorbergh (2014)

Chen, Roll, and Ross: Interpretable Factors

- Industrial Production Growth (*MP*)
- Change in Expected Inflation (*DEI*)
- Unexpected Inflation (*UI*)
- Default Premium (*UPR*)
- Term Premium (*UTS*)

CRR Factors Work Across Asset Classes!



Success?

Models	N factors	GRS - stat
Rm	1	4.08
Rm+VAL+MOM (Asness, Moskowitz, and Pedersen)	3	2.84
Five global macro factors (this paper)	5	2.78

Dark Side of APT

- Common factor structure can be a gift...
- ... but also a curse (e.g., Lewellen, Nagel, and Shanken (2010))

Do we need all five macro factors?

Models	N factors	GRS - stat
Rm	1	4.08
Rm+VAL+MOM (AsnessMoskowitzPedersen)	3	2.84
Five global macro factors (CooperMitrachePristley)	5	2.78
Four global macro factors (minus Ind prod)	4	3.05
Four global macro factors (minus Unexp Inflation)	4	2.84
Four global macro factors (minus Exp Inflation)	4	3.08
Four global macro factors (minus Global Term Prem)	4	4.23
Four global macro factors (minus US default spread)	4	2.82
Three global macro factors (IP, Unexp+Exp inflation)	3	4.22
Three global macro factors (IP, Unexp inflation + UTS)	3	2.90

Alternative Factors?

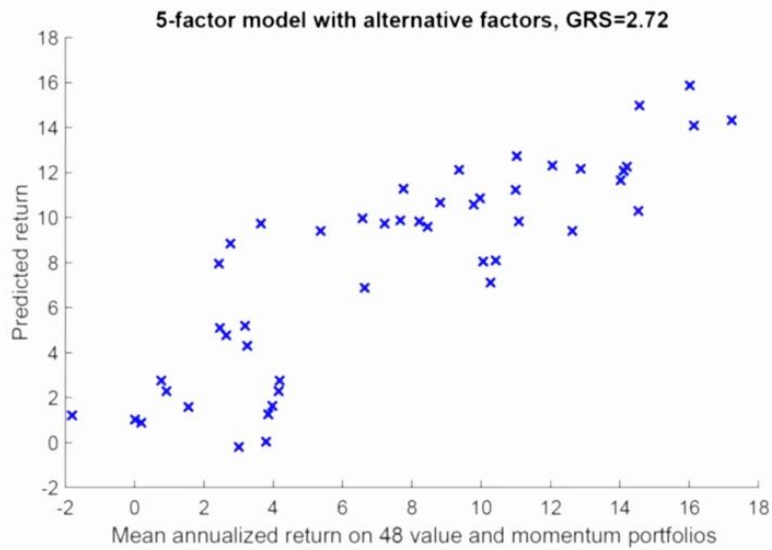
How About the Following (all growth rates):

- Assets of commercial banks
- Civilian labor force
- Passenger car registrations
- Loans and leases in bank credit
- New housing permits

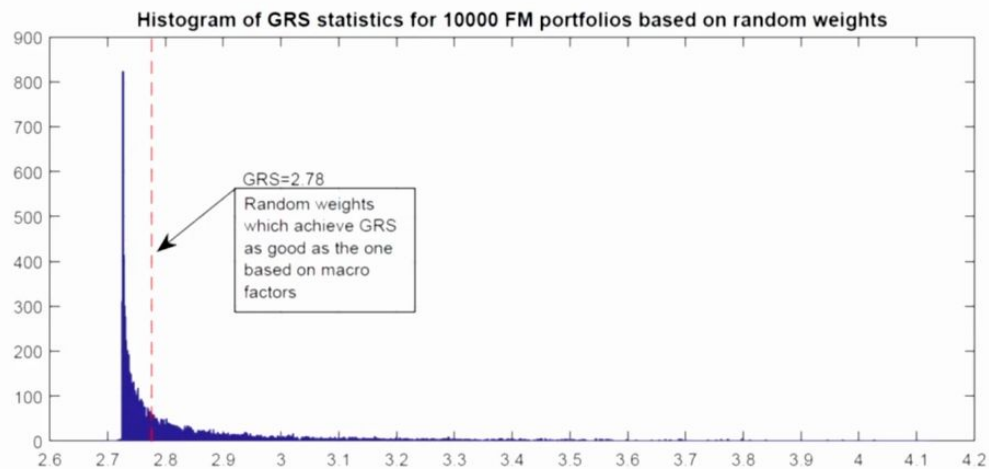
Alternative Factors

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Four global macro factors (minus US default spread)	4	2.82
Three global macro factors (IP, Unexp+Exp inflation)	3	4.22
Three global macro factors (IP, Unexp inflation + UTS)	3	2.90
High, mid, low val and mom, global	6	2.71
High, low val and mom, global	4	2.73
Alternative US macro factors	5	2.72

Alternative Factors Work Too!



Random Factor-Mimicking Portfolios



Concluding Thoughts

- Key result: global macro factors related to global value and momentum returns!
- Too many degrees of freedom?
- Which factors most important? Why?